

Summary

Founding Product Engineer / AI Engineer building **production AI systems that turn messy real-world data into structured intelligence**. Experienced across **RAG, multi-agent workflows, backend APIs, multi-tenant authentication, usage-based billing, retrieval systems, observability, and real-time user-facing products**. Strong at translating ambiguous business workflows into scalable systems, owning product execution, and shipping quickly in startup environments.

Skills

AI Agents & LLM Systems: Multi-Agent Workflows, LangGraph, LangChain, RAG, Crawl4AI, Prompt Orchestration, Tool Use / Function Calling, Agent Memory, Guardrails, Evaluation, Langfuse, GoogleADK, ClaudeSDK

Backend Engineering: FastAPI, REST APIs, Microservices, AsyncIO, Flask, React/Node.js, TypeScript, Auth0, Stripe, Docker, CI/CD, Web Server Deployment

Data, & Analytics: PostgreSQL, Cloud SQL, MongoDB, Vector Search, Pinecone, FAISS, ETL Pipelines, Hybrid Retrieval, Sentry, PostHog

Machine Learning: PyTorch, TensorFlow, XGBoost, YOLO, Computer Vision, Model Optimization, MLflow

Languages & Cloud: Python, TypeScript, SQL, Go, C++, Rust, GCP, AWS, Azure, CUDA

Experience

AI Software Engineer (Forward Deployed) — ProspectIntel Jan 2026 – Present

- Spearheaded **0→1 development** of a production AI sales intelligence platform, translating messy market data into structured insights using multi-agent workflows, schema-constrained LLMs, and end-to-end full-stack execution.
- Engineered **dynamic RAG pipelines** using Crawl4AI and hybrid PostgreSQL retrieval, enabling agents to reason over fresh, real-world context rather than static database records, improving output reliability from **70% to 95%** and reducing latency by **2–3×**.
- Architected end-to-end agent orchestration, lazy-loading inference and streaming LLM responses to support concurrent users and cut perceived latency by 40%, and exposed the platform's intelligence to external AI clients via a standalone MCP (Model Context Protocol) server (FastMCP, streamable-HTTP, Auth0 OIDC/OAuth).
- Developed **enterprise B2B infrastructure** featuring Auth0 multi-tenant onboarding and strict organization-aware access controls across the frontend, backend, and database state with Row Level Security.
- Built **custom token metering and monetization logic**, integrating Stripe usage-based billing across LLMs, web research, and external APIs to enable precise cost attribution for AI-heavy workflows.
- Established **comprehensive observability pipelines** using Langfuse, Sentry, and PostHog to track prompt performance, monitor production errors, and optimize AI behavior, reducing overall token consumption by **35%**.

AI Software Developer — Bruise @ GMU Jun 2025 – Present

- Engineered a **HIPAA-compliant full-stack AI platform** using FastAPI and React/Next.js for real-time clinical workflows, deployed across universities (USA & Africa) and used by **50+ clinicians and 10+ researchers**.
- Led end-to-end infrastructure by independently deploying on a **fresh production server** and architecting a highly reliable system that separates APIs, inference, and data pipelines into **modular services**.
- Built a **Python HTTP gateway** bridging a Go backend with a dedicated **YOLO-based AI inference microservice**, enabling asynchronous cross-language communication and offloading heavy compute.
- Designed robust data pipelines to extract, clean, and normalize **Base64-encoded medical images and metadata** from PostgreSQL for downstream clinical analysis and AI routing.
- Built **LLM-powered clinical summarization pipelines** with voice agents and agentic orchestration that seamlessly synthesize structured injury data and clinical context into actionable, highly usable reports for medical researchers.
- Integrated **3D spatial mapping via Unity** with backend services, significantly enhancing the visualization, usability, and interpretation of complex clinical injury data.

Software Engineer (ML) — Geoweaver (NASA / ESIP) Feb 2024 – Jan 2025

- Built and productionized **scalable ML pipelines** for TB-scale geospatial datasets including snow water estimation, wildfire detection, and ozone analysis using Python and PostgreSQL.
- Executed and optimized workflows on **HPC systems**, leveraging distributed processing to improve compute efficiency by **20%** and accelerate large-scale scientific data analysis.
- Designed and integrated **multi-database support** and contributed to an in-app **code editor**, enabling flexible data access and improving developer workflows within the platform.
- Improved **UI responsiveness and feature experience**, reducing latency and contributing to a **25% increase in monthly active users**.
- Mentored **4 interns** in ML experimentation and system design, co-authoring a peer-reviewed research publication.

AI/ML Engineer — Tech Solutions Nov 2021 – Jul 2023

- Built **real-time fraud detection systems** using Flask APIs and XGBoost models, enabling low-latency transaction risk scoring and reducing high-risk exposure by **15%**.
- Designed and deployed **scalable backend services and ETL pipelines** processing **10M+ financial transactions/month**, powering automated risk analysis and decision workflows.
- Developed **production-grade APIs** and asynchronous processing pipelines to support real-time business operations and explainable ML-driven decisions.
- Worked with **AWS infrastructure** including EC2, S3, CloudWatch, and Docker-based deployments to support scalable model serving, monitoring, and reliable backend operations.
- Implemented **CI/CD and MLOps workflows** using Docker, GitHub Actions, and MLflow, improving deployment reliability and reducing iteration time by **30%**.

Education

George Mason University, Fairfax, VA — M.S. Computer Science 2023–2025

GGG Indraprastha University, Delhi, India — B.Tech. Information Technology 2019–2023